Content Vs Form: What Drives Differences in Test Scores Across Different Socioeconomic Status

1 Introduction

For many of us, thinking and articulating a thought are two distinct concepts. Sometimes, we may have a clear mental image of an idea, but when we try to express it aloud and convey it to others, we encounter additional challenges. This can be especially important in situations where we are being evaluated. We may possess a very good idea and a coherent answer, but we might be unable to express it in writing in a manner that convinces the evaluator that our argument is coherent and addresses all relevant points. This is especially true in cases where the evaluator and students do not share a common background, and the evaluator is facing mental constraints or implicit biases that prevent them from fully appreciating the ideas conveyed in the text.

The ability to clearly articulate ideas in writing and target our writing to evaluators or specific audiences is a skill that is cultivated over time. It's influenced by a variety of factors, including educational background, access to resources, and socio-economic status. These factors can create disparities in how students from different social groups express their thoughts. Recognizing and addressing these disparities is crucial for educators. It helps them understand where to focus their efforts, whether on improving writing skills or ensuring that students' ideas are evaluated fairly, regardless of their presentation.

From the perspective of educational fairness, it becomes important to consider whether the evaluation of written work should prioritize the originality and validity of the ideas rather than the fluency of their expression. If the goal is to assess the substance of a student's understanding and creativity, educators might need to adjust their grading strategies to account for differences in writing style that are influenced by external factors beyond a student's control. This approach could involve offering additional support to students who face challenges in articulating their thoughts due to their backgrounds, thereby leveling the playing field. Such measures not only promote equity in educational outcomes but also encourage a more inclusive and diverse range of perspectives in academic discourse.

We therefore in this paper explroe the source of the gap in scores. We do so by using a OXCAC style decomposition, decomposing the gap in scores

MAin challange is is to seperate the We see that the in general information in style and xxx - therefore xcxcv

To overcome this we propose a new method to do so. We rewrite the articles on XXx

We document in our sample that much of the variation in scores stems from differences in content. Approximately XX% of the gap is explained by content differences, xx% by differences in style, and the remainder by unexplained variation associated with the scoring functions applied to low- and high-SES individuals. The fact that xx% of the score gap is driven by style is substantial, suggesting that improving articulation could meaningfully reduce disparities across groups.

Although ideas and reasononig may stem from idea, articualtion and stnadarization canxx.vxv

Related Literature It's well-documented that essay scoring differences across social groups are significant and persistent [**?**]. Numerous studies have examined the causes of these scoring disparities, mainly focusing on the correlation between writing style, structure, and quality (including grammar) and the eventual scores [**?**]. While these studies suggest some relationships, they fail to control for the context in which the writing was delivered. Other research has investigated biases in automatic scoring algorithms [**?**], revealing varying biases depending on the fairness measures applied.

Our method for measuring disparities aligns with recent trends in economics and computer science research. These trends concentrate on disparities in measured outcomes after accounting for a specific variable of interest [**?**, **?**]. For instance, [**?**] examined disparity in judicial treatment by analyzing variations in bail decisions for individuals with similar probabilities of reoffending or failing to appear in court. [**?**] investigated employment probability disparities among workers of equal productivity but from different groups. Such approaches underscore the normative aspect of measuring disparities across social groups by pinpointing variables that warrant equal treatment. Our approach is slightly different, aiming to identify not only differential impacts but also the driving factors behind these disparities. This method aligns with more traditional economic approach that deconstructs observed gaps by creating "counterfactual distributions," similar to our approach here ([**?**]), where we fix one variable and modify the conditional distribution to create the counterfactual.

Finally, our approach resonates with aspects of the explainable AI literature, which aims to understand algorithmic predictions by altering input components. For example, in text analysis, some studies, akin to ours, have proposed modifying parts of the text (such as changing adjectives from positive to negative) to affect sentiment prediction [**?**, **?**]. Our method is more conservative, providing decomposition analysis rather than claiming causality without additional assumptions, like the random assignment of tests to graders. Other studies have explored input modification to gain insights into algorithms or underlying distributions. [**?**], for instance, investigated factors influencing judicial bail decisions by examining why pictures of defendants are significant predictors for judges' decisions. They accomplished this by randomly generating a picture and morphing it with minimal facial changes that significantly impact bail approval likelihood. This process results in two pictures that are substantially different in terms of release probability, yet similar in appearance. Analyzing the differences between these two pictures aids in formulating hypotheses

about what contributes to the considerable variance in bail decisions.

2 Framework

We observe essays indexed by $i = 1, \ldots, N$, each with an original rendering (k=0) and $K \geq 1$ content-preserving rewrites ($k = 1, \ldots, K$). Let $G_i \in \{H, L\}$ denote group (SES). For each *scoring concept* or *ranker* $m \in \{H, L\}$, a learned deterministic scoring function $S^{(m)}$ maps a rendered text x_{ik} to a numeric score:

$$s_{ik}^{(m)} \;=\; S^{(m)}(x_{ik}) \in \mathbb{R}.$$

Write C_i for the (latent) *content* of essay i; rewrites preserve C_i by design. Let R_{ik} denote the phrasing/style realization of version k for paper i. We make the following assumption on the scoring function.

Assumption 1 (Separable content and style)**.** *For any scorer m the essay scoring is separable in content and style:*

$$s_{ik}^{(m)} = \underbrace{\theta_m(C_i)}_{\textit{Content index under } m} + \underbrace{\rho_m(R_{ik})}_{\textit{Style component under } m}. \tag{1}$$

This assumption on the ranker implies that scoring varies additively with the content and with the style, where θ_m and ρ_m are mappings from content and style, respectively, to the score under m. Notice that we allow for different rankers to weight content and style differently. For example, rankers for one group of students may be more responsive to style and less responsive to the actual content of the answer.

We also make the following assumptions on the scorer function.

Assumption 2 (Mean Independence)**.** *Let $P^*(.|C)$ be the distribution of rewrites of concept C. Then for any scorer m and writing-style distribution,*

$$\mathbb{E}_{R\sim P^*(\cdot|C_i)}\big[\rho_m(R)\big] \;=\; 0.$$

Assumption 3 (LLM Mean Equivalence)**.** *For any scorer m and content C_i,*

$$\mathbb{E}_{R\sim P^*(\cdot|C_i)}\big[S^{(m)}(C_i, R)\big] \;-\; \mathbb{E}_{R\sim P^{\mathrm{LLM}}(\cdot|C_i)}\big[S^{(m)}(C_i, R)\big] \;=\; \kappa_m.$$

Finally we also impose the assumption on the quality of rewrites.

Assumption 4 (Content Fidelity)**.** *All rewrites preserve content. C_i is constant across k for each i.*

Discussion. Assumption 2 is standard in econometrics. It implies that variation in scores due to writing style is centered around the underlying content score, so that some styles may raise scores while others may lower them. Assumption 3 is stronger: it requires that, conditional on content,

the mean score distribution generated by the LLM coincides with the mean over all possible human rewrites of the same content, up to a content-independent constant. Finally, Assumption 4 is an assumption on the design, and implies that we can create variation in style holding content fixed, and that the LLM rewrites do a good job at preserving content while changing the style.

Remark 1. *Assumption* **??** *can be interpreted in two ways. Positively, it states that the mean of the score distribution is centered on the fixed component. Normatively, it serves as a definition: even if the scores are not centered on the true mean, any systematic effect of paraphrasing on the mean score should be absorbed into the constant.*

2.1 When the decomposition is Causal

This decomposition is causal under the assumptions

3 Decomposition of the observed gap

Suppose the policy grades H originals with $S^{(H)}$ and L originals with $S^{(L)}$. The observed gap is

$$\Delta_{\text{obs}} := \mathbb{E}_H[s_{i0}^{(H)}] - \mathbb{E}_L[s_{i0}^{(L)}].$$

We suggest decomposing the gap as follows. Let the group means $A_G^{(m)} := \mathbb{E}_G[\widehat{\alpha}^{(m)}]$, $U_G^{(m)} := \mathbb{E}_G[\widehat{u}_{i0}^{(m)}]$, where the fixed effects $\widehat{\alpha}_i^{(m)}$ and original deviations $\widehat{u}_{i0}^{(m)}$ are defined in the estimation section. The following identity holds:

$$\begin{aligned}\Delta_{\text{obs}} &= \mathbb{E}_H\big[s_{i0}^{(H)}\big] - \mathbb{E}_L\big[s_{i0}^{(H)}\big] \;+\; \mathbb{E}_L\big[s_{i0}^{(H)}\big] - \mathbb{E}_L\big[s_{i0}^{(L)}\big] \\ &= \underbrace{(A_H^{(H)} - A_L^{(H)})}_{\text{content under } S^{(H)}} + \underbrace{(U_H^{(H)} - U_L^{(H)})}_{\text{style (original) under } S^{(H)}} + \underbrace{\mathbb{E}_L\big[s_{i0}^{(H)} - s_{i0}^{(L)}\big]}_{\text{Ranker Tilting}}. \end{aligned} \tag{2}$$

Interpretation. The first bracket tells us how much of the difference in scores is driven by differences in content quality across the two groups, when they are ranked by the High-SES scorer. The second bracket measures how much of the gap is driven by difference in the writing style of the two groups, if they were ranked by the High-SES scorer. Finally, the last component captures how much of the difference is driven by the rankers themselves (measured on the same L texts).

Remark 2. *Although Assumption 3 requires the mean score across concepts to be captured exactly by the LLM, for the decomposition we can slightly relax this and allow for a* systemic constant *difference between the theoretical score mean (conditional on content) and the realized LLM mean. This constant difference cancels in the within-scorer FE differences and thus does not affect* (2)*.*

3.1 Estimation and implementation

Data construction. For each essay, generate K content-faithful paraphrases using diverse prompts to approximate a broad phrasing distribution. Score *every* version with both rankers to obtain

$s_{ik}^{(H)}, s_{ik}^{(L)}$.

Fixed effects (projection, not raw means). For each scorer m, estimate the fixed effects model

$$s_{i,k} = \alpha_i + \gamma_k + \epsilon_{i,k}.$$

Here, α_i captures the fixed component—the content—while γ_k and the residuals capture the style components. Define the original deviation

$$\widehat{u}_{i0}^{(m)} \;:=\; s_{i0}^{(m)} \;-\; \widehat{\alpha}_i^{(m)}. \tag{3}$$

Notice that γ_k goes into $\widehat{u}_{i0}^{(m)}$, as γ_k is driven by the writing style.

Decomposition. Compute $\widehat{\alpha}_i^{(m)}$ and $\widehat{u}_{i0}^{(m)}$; form group means $A_G^{(m)}, U_G^{(m)}$; plug into (2).

4 Data

The persuade dataset Our measures for style Embeddign - We represent each text in our corpus using sentence-level embeddings derived from the [CLS] token of a pre-trained BERT model (Devlin et al., 2019). Specifically, for each text segment, we extract the contextualized embedding of the [CLS] token from the final layer, which is commonly interpreted as a summary representation of the entire sequence. This approach allows us to encode the semantic and conceptual content of each text into a fixed-dimensional vector that captures its underlying "idea" in a meaningful way. Prior work has shown that BERT embeddings effectively encode higher-level abstractions such as topic, intent, and relational meaning, beyond surface-level lexical similarity (Reimers Gurevych, 2019; Ethayarajh, 2019). Thus, using the [CLS] representation provides a principled method for mapping texts into a semantic space where distances reflect conceptual relatedness rather than merely word overlap.

5 Descriptive Statistics

5.1 Descriptives

The sample, the score gap share of XX, some charchtristics, style predictions - discuss those Descriptives on the essays (number of words, number of sentences, reading level,

5.2 Topics are slightly different

the low dim estimation, predicting histogram

5.3 Content and Style are highly intertwind

- similar predicted scores (show histogram and correlations between predicting with emb + predicting with style) - Do the R2 square analysis - show the r sqaured

- Figure 12 shows that low-SES students tend to write in styles more strongly associated with lower scores, whereas high-SES students are more evenly distributed across the support. Figure 11 reveals similar patterns. Together, these figures demonstrate that ideas and writing style are closely intertwined—information about one provides insight into the other. This relationship motivates our rewriting approach in Section XX. Additional results are presented in Appendix XX, where we show XXX.

- Figure XX shows that

6 Decomposition Results

6.1 Scorer Function

How the score function is doing - confusion matrix, and grey graph

6.2 Rewrites

6.3 Decomposition Results + histogram of fixed effect and style

MAIN REUSLTS, second decomoposiotn approach

6.4 Robustness

- try different rewrites approaches, omit the original from the fixed effect estimation,

7 Conclusions

8 Appendix

8.1 Approach II: pooled FE with common content (assumes ranker additivity).

Stack both rankers and estimate

$$s_{ik}^{(m)} = \mu + \alpha_i + \tau_m + r_{ik}^{(m)}, \qquad \sum_i \alpha_i = \sum_m \tau_m = 0, \tag{4}$$

i.e., a *common* paper effect α_i across rankers plus a ranker FE τ_m. Then, for originals,

$$\boxed{\Delta_{\text{obs}} = \underbrace{\big(\mathbb{E}_H[\widehat{\alpha}_i] - \mathbb{E}_L[\widehat{\alpha}_i]\big)}_{\text{content (common FE)}} + \underbrace{(\widehat{\tau}_H - \widehat{\tau}_L)}_{\text{ranker (systemic)}} + \underbrace{\big(\mathbb{E}_H[\widehat{r}_{i0}^{(H)}] - \mathbb{E}_L[\widehat{r}_{i0}^{(L)}]\big)}_{\text{style (non-separable)}}}. \tag{5}$$

Here *all* ranker differences that are paper-invariant collapse into $\tau_H - \tau_L$; departures live in the residuals $r_{ik}^{(m)}$. The key additional restriction is *no paper×ranker interaction in levels.* It is testable by augmenting (4) with interaction dummies $\delta_i^{(m)}$ (sum-to-zero over m) and testing $\delta_i^{(m)} = 0$ jointly.

8.2 Baseline Decomposition

We can relax the separability assumption, if we are willing to accept a "neutral" baseline decomposition. This decomposition is more informative is we want to think of our decomposition from a more policy perspective - i.e. assume we let students use GPT to improve their writing - how would that affect the gap?

We start by defining new notation. For group $G \in \{H, L\}$, let

$$\mu_G^{S,\text{orig}} := \mathbb{E}\big[S(\text{original text}) \mid G\big], \qquad \mu_G^{S,\text{neu}} := \mathbb{E}\big[S(T(\text{original text})) \mid G\big],$$

where T rewrites to GPT-style ("neutral"). Scoring functions: $S^{(H)}$ and $S^{(L)}$.

The observed score gap is given by the following expression:

$$\Delta_{\text{obs}} \;=\; \mu_H^{S^{(H)},\text{orig}} \;-\; \mu_L^{S^{(L)},\text{orig}}.$$

We now derive the decomposition term.

Step 1: Introduce the scoring-function *tilt* (L only)

Add and subtract $\mu_L^{S^{(H)},\text{orig}}$:

$$\Delta_{\text{obs}} = \big(\mu_H^{S^{(H)},\text{orig}} - \mu_L^{S^{(H)},\text{orig}}\big) \;+\; \underbrace{\big(\mu_L^{S^{(H)},\text{orig}} - \mu_L^{S^{(L)},\text{orig}}\big)}_{\text{Scoring-function tilt (vs. } S^{(H)})} .$$

Step 2: Split the same-scorer difference into *content* and *style*

Now add and subtract the neutralized means *under the same scorer* $S^{(H)}$:

$$\mu_H^{S^{(H)},\text{orig}} - \mu_L^{S^{(H)},\text{orig}} = \underbrace{\big(\mu_H^{S^{(H)},\text{neu}} - \mu_L^{S^{(H)},\text{neu}}\big)}_{\text{Content}^{(H)}} + \underbrace{\big(\mu_H^{S^{(H)},\text{orig}} - \mu_H^{S^{(H)},\text{neu}}\big)}_{\text{H style premium under } S^{(H)}} - \underbrace{\big(\mu_L^{S^{(H)},\text{orig}} - \mu_L^{S^{(H)},\text{neu}}\big)}_{\text{L style premium under } S^{(H)}}.$$

Taking the difference of the two blue terms gives the net

$$\text{Style}^{(H)} \;=\; \big(\mu_H^{S^{(H)},\text{orig}} - \mu_H^{S^{(H)},\text{neu}}\big) \;-\; \big(\mu_L^{S^{(H)},\text{orig}} - \mu_L^{S^{(H)},\text{neu}}\big).$$

He the style can be thought of as the premium of the high and low SES writing style compare to the neutral level.

Step 3: Combine

Putting Steps 1 and 2 together:

$$\boxed{\Delta_{\text{obs}} = \underbrace{\left(\mu_H^{S^{(H)},\text{neu}} - \mu_L^{S^{(H)},\text{neu}}\right)}_{\text{Content}^{(H)}} + \underbrace{\left[\left(\mu_H^{S^{(H)},\text{orig}} - \mu_H^{S^{(H)},\text{neu}}\right) - \left(\mu_L^{S^{(H)},\text{orig}} - \mu_L^{S^{(H)},\text{neu}}\right)\right]}_{\text{Style}^{(H)}} + \underbrace{\left(\mu_L^{S^{(H)},\text{orig}} - \mu_L^{S^{(L)},\text{orig}}\right)}_{\text{Scoring-function tilt (vs. } S^{(H)})}}$$

8.3 Robustness

8.4 Sensitivity of results to rewrite

9 Figures

9.1 Descriptive Statistics

	High SES	Low SES	All
Scores	3.65	3.23	3.51
Women	3.78	3.33	3.64
Minorities	3.70	3.22	3.49
Average Readability Score	9.64	9.83	9.70
Average Number of Words	445.03	426.71	439.17

Table 1: Descriptive Statistics (average scores across SES groups by gender, race, and text features)

figures/real_scores.png

Figure 1: Distribution of low SES scores and High SES Scores - real

figures/embeddings_UMAP.png

Figure 2: 2d embedding seperate by High SES vs Low SES - UMAP

figures/t-sne-kde-test.png

Figure 3: 2d embedding seperate by High SES vs Low SES - t-SNE

9.2 Ranker

figures/ranker_high_SES.png

Figure 4: Confusion matrix - high ses

figures/ranker_low_SES.png

Figure 5: Confusion matrix - low Ses

figures/predicted_scores_high.png

Figure 6: Distribution of predicted scores - high vs low SES - High Scorer

figures/predicted_scores_low.png

Figure 7: Distribution of predicted scores - high vs low SES - Low Scorer

figures/binned_low.png

Figure 8: Binned Values - predicted valeu vs estimated mean - Low SES

figures/binned_high.png

Figure 9: Binned Values - predicted valeu vs estimated mean - high ranker

figures/explained_variability.png

Figure 10: Share of explained variability - Embedding Vs style Vs raw

Note: high overlap between embedding and style - hard to distinguish between the two - ideas determine style

figures/predicted_scores_style.png

Figure 11: Distribution of predicted score using only style - low vs high ses

figures/predicted_scores_emb.png

Figure 12: Distribution of predicted score using only embedding - low vs high ses

figures/predicted_scores_std.png

Figure 13: high SES predicted Valeu vs Low SES predicted value

9.3 Rewrites

unit_did_hist_unemp_share.pdf

Figure 14: Examples of rewrites - original vs Rewritten

figures/predicted_scores_k_high_density.png

Figure 15: Distribution of rewrites predicted score - high ranker

figures/predicted_scores_k_low_density.png

Figure 16: Distribution of rewrites predicted score - low ranker

figures/predicted_mean_k.png

Figure 17: Bar graph of mean prediction for each k X SES

figures/predicted_score_style_fixed_high_density.png

Figure 18: Distribution of rewrites predicted score based only on style - high ranker

10 Main Results

figures/fixed_effect_distribution.png

Figure 19: Fixed Effect distribution - (empircal Bayes Correction)

figures/style_distribution.png

Figure 20: Style distribution - (empircal Bayes Correction)

figures/new_style/decomp_shares_all.png

Figure 21: Decomposition Shares (All, By Gender, by Race, By question